How to be Pessimistic: Choquet Risk and Porfolio Optimization

Gilbert W. Bassett, Jr.
University of Illinois, Chicago
Roger Koenker
University of Illinois, Urbana-Champaign
Gregory Kordas
University of Pennsylvania

4th International Conference on $\mathcal{L}_1$ and Related Methods:
Neuchatel, August 2002

St. Petersburg Paradox

What would you be willing to pay to play the game:

$$G = \{\text{pay: } p \quad \text{win: } 2^n \text{ with probability } 2^{-n}\}$$

St. Petersburg Paradox

What would you be willing to pay to play the game:

$$G = \{\text{pay: } p \quad \text{win: } 2^n \text{ with probability } 2^{-n}\}$$

Bernoulli's observation that even though the expected payoff was infinite, the gambler who maximized logarithmic utility would only pay a finite value to play. For example, given initial weath 100,000 Roubles, our gambler would pay only 17 Roubles and 55 kopecks.

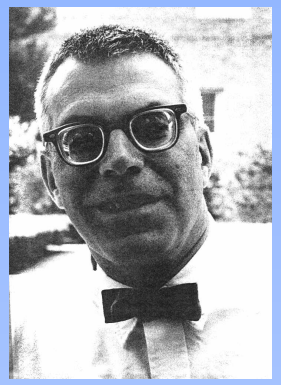
Expected Utility

To decide between two real valued gambles

$$X \sim F \quad \text{and} \quad Y \sim G$$

we choose X over Y if

$$Eu(X) = \int u(x)dF(x) \geq \int u(y)dG(y) = Eu(Y)$$



What is Risk?

In the expected utility theory risk is in effect an attribute of the utility function:

Risk Neutrality	$\Rightarrow$	$u(x) \sim$ affine
Risk Aversion	$\Rightarrow$	$u(x) \sim$ concave
Risk Attraction	$\Rightarrow$	$u(x) \sim$ convex

What is Risk?

In the expected utility theory risk is in effect an attribute of the utility function:

Risk Neutrality	$\Rightarrow$	$u(x) \sim$ affine
Risk Aversion	$\Rightarrow$	$u(x) \sim$ concave
Risk Attraction	$\Rightarrow$	$u(x) \sim$ convex

Locally, the risk premium, i.e. the amount one is willing to pay to accept a zero mean risk, X, is

$$\pi(w, X) = \tfrac{1}{2} A(w) V(X)$$

where $A(w) = -u''(w)/u'(w)$ is the Arrow-Pratt coefficient of absolute risk aversion and $V(X)$ is the variance of X.

What is Risk?

In the expected utility theory risk is in effect an attribute of the utility function:

Risk Neutrality	$\Rightarrow$	$u(x) \sim$ affine
Risk Aversion	$\Rightarrow$	$u(x) \sim$ concave
Risk Attraction	$\Rightarrow$	$u(x) \sim$ convex

Locally, the risk premium, i.e. the amount one is willing to pay to accept a zero mean risk, X, is

$$\pi(w, X) = \tfrac{1}{2} A(w) V(X)$$

where $A(w) = -u''(w)/u'(w)$ is the Arrow-Pratt coefficient of absolute risk aversion and $V(X)$ is the variance of X. The variance of X?

A Little Risk Aversion is a Dangerous Thing

Would you accept the gamble:

$$G_1 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$110 \\ \text{lose } \$100 \end{array} \right.$$

A Little Risk Aversion is a Dangerous Thing

Would you accept the gamble:

$$G_1 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$110 \\ \text{lose } \$100 \end{array} \right.$$

Suppose you say "no", then what about the gamble:

A Little Risk Aversion is a Dangerous Thing

Would you accept the gamble:

$$G_1 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$110 \\ \text{lose } \$100 \end{array} \right.$$

Suppose you say "no", then what about the gamble:

$$G_2 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$700,000 \\ \text{lose } \$1,000 \end{array} \right.$$

A Little Risk Aversion is a Dangerous Thing

Would you accept the gamble:

$$G_1 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$110 \\ \text{lose } \$100 \end{array} \right.$$

Suppose you say "no", then what about the gamble:

$$G_2 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$700,000 \\ \text{lose } \$1,000 \end{array} \right.$$

If you say "no" to G_1 for any initial wealth up to \$300,000, then you must also say "no" to G_2.

A Little Risk Aversion is a Dangerous Thing

Would you accept the gamble:

$$G_1 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$110 \\ \text{lose } \$100 \end{array} \right.$$

Suppose you say "no", then what about the gamble:

$$G_2 \qquad 50 - 50 \quad \left\langle \begin{array}{l} \text{win } \$700,000 \\ \text{lose } \$1,000 \end{array} \right.$$

If you say "no" to G_1 for any initial wealth up to \$300,000, then you must also say "no" to G_2.

Moral A little local risk aversion over small gambles implies implausibly large risk aversion over large gambles. Reference: Rabin (2000)

Are Bicycle Messengers Risk Averse?

When Veloblitz and Flash Messengers from Zurich were confronted with the bet:

$$50 - 50 \quad \left\langle \begin{array}{l} \text{win } 8\,CHF \\ \text{lose } 5\,CHF \end{array} \right.$$

Are Bicycle Messengers Risk Averse?

When Veloblitz and Flash Messengers from Zurich were confronted with the bet:

$$50 - 50 \quad \left\langle \begin{array}{l} \text{win } 8\,CHF \\ \text{lose } 5\,CHF \end{array} \right.$$

54% rejected the bet.

Reference: Fehr and Götte (2002)

Choquet Expected Utility

Among the many proposals offered to extend expected utility theory the most attractive (to us) replaces

$$E_F u(X) = \int_0^1 u(F^{-1}(t))dt \geq \int_0^1 u(G^{-1}(t))dt = E_G u(Y)$$

with

Choquet Expected Utility

Among the many proposals offered to extend expected utility theory the most attractive (to us) replaces

$$E_F u(X) = \int_0^1 u(F^{-1}(t))dt \geq \int_0^1 u(G^{-1}(t))dt = E_G u(Y)$$

with

$$E_{\nu,F} u(X) = \int_0^1 u(F^{-1}(t))d\nu(t) \geq \int_0^1 u(G^{-1}(t))d\nu(t) = E_{\nu,G} u(Y)$$

Choquet Expected Utility

Among the many proposals offered to extend expected utility theory the most attractive (to us) replaces

$$E_F u(X) = \int_0^1 u(F^{-1}(t))dt \geq \int_0^1 u(G^{-1}(t))dt = E_G u(Y)$$

with

$$E_{\nu,F} u(X) = \int_0^1 u(F^{-1}(t))d\nu(t) \geq \int_0^1 u(G^{-1}(t))d\nu(t) = E_{\nu,G} u(Y)$$

The measure ν distorts the probability assessments after ordering the outcomes. This rank dependent form of expected utility has been pioneered by Quiggin (1981), Schmeidler (1989), Walker (1989) and Dennenberg (1990).

Pessimism

The simplest form of Choquet expected utility is based on the "distortion"

$$\nu_\alpha(t) = \min\{t/\alpha, 1\}$$

so

$$E_{\nu_\alpha,F} u(X) = \alpha^{-1} \int_0^\alpha u(F^{-1}(t))dt$$

This increases the probability of the proportion α of least favorable events, and totally discounts the probability of the $1-\alpha$ most favorable events.

Pessimism

The simplest form of Choquet expected utility is based on the "distortion"

$$\nu_\alpha(t) = \min\{t/\alpha, 1\}$$

so

$$E_{\nu_\alpha,F} u(X) = \alpha^{-1} \int_0^\alpha u(F^{-1}(t))dt$$

This increases the probability of the proportion α of least favorable events, and totally discounts the probability of the $1-\alpha$ most favorable events.

Expect the worst – and you won't be disappointed.

Savage on Pessimism

I have, at least once heard it objected against the personalistic view of probability that, according to that view, two people might be of different opinions, according as one is pessimistic and the other optimistic. I am not sure what position I would take in abstract discussion of whether that alleged property of personalistic views would be objectionable,

Savage on Pessimism

I have, at least once heard it objected against the personalistic view of probability that, according to that view, two people might be of different opinions, according as one is pessimistic and the other optimistic. I am not sure what position I would take in abstract discussion of whether that alleged property of personalistic views would be objectionable, but I think it is clear from the formal definition of qualitative probability that the particular personalistic view sponsored here does not leave room for optimism and pessimism, however these traits may be interpreted, to play any role in the person's judgement of probabilities. (Savage(1954, p. 68)

Coherent Risk

Definition (Artzner, Delbaen, Eber and Heath (1999)) For real valued random variables $X \in \mathcal{X}$ on $(\Omega, \mathcal{A})$ a mapping $\varrho : \mathcal{X} \to \mathcal{R}$ is called a coherent risk measure if,

1. Monotone: $X, Y \in \mathcal{X}$, with $X \leq Y \Rightarrow \varrho(X) \geq \varrho(Y)$.
2. Subadditive: $X, Y, X + Y \in \mathcal{X}$, $\Rightarrow \varrho(X + Y) \leq \varrho(X) + \varrho(Y)$.
3. Linearly Homogeneous: For all $\lambda \geq 0$ and $X \in \mathcal{X}$, $\varrho(\lambda X) = \lambda \varrho(X)$.
4. Translation Invariant: For all $\lambda \in \mathcal{R}$ and $X \in \mathcal{X}$, $\varrho(\lambda + X) = \varrho(X) - \lambda$.

Coherent Risk

Definition (Artzner, Delbaen, Eber and Heath (1999)) For real valued random variables $X \in \mathcal{X}$ on $(\Omega, \mathcal{A})$ a mapping $\varrho : \mathcal{X} \to \mathcal{R}$ is called a coherent risk measure if,

1. Monotone: $X, Y \in \mathcal{X}$, with $X \leq Y \Rightarrow \varrho(X) \geq \varrho(Y)$.
2. Subadditive: $X, Y, X + Y \in \mathcal{X}$, $\Rightarrow \varrho(X + Y) \leq \varrho(X) + \varrho(Y)$.
3. Linearly Homogeneous: For all $\lambda \geq 0$ and $X \in \mathcal{X}$, $\varrho(\lambda X) = \lambda \varrho(X)$.
4. Translation Invariant: For all $\lambda \in \mathcal{R}$ and $X \in \mathcal{X}$, $\varrho(\lambda + X) = \varrho(X) - \lambda$.

Many conventional measures of risks including those based on standard deviation are ruled out, as are quantile based measures like "value at risk."

Choquet α-Risk

The leading example of a coherent risk measure is

$$\varrho_{\nu_\alpha}(X) = -\alpha^{-1} \int_0^\alpha F^{-1}(t)dt$$

Variants of this risk measure have been introduced under several names

- Expected shortfall (Acerbi and Tasche (2002))
- Conditional VaR (Rockafellar and Uryasev (2000))
- Tail conditional expectation (Artzner, et al (1999)).

Note that $\varrho_{\nu_\alpha}(X) = -E_{\nu_\alpha,F}(X)$, so Choquet α-risk is just negative Choquet expected utility with distortion function ν_α.

Pessimistic Risk Measures

Definition A risk measure ϱ will be called pessimistic if, for some probability measure ν on [0, 1]

$$\varrho(X) = \int_0^1 \varrho_{\nu_\alpha}(X) d\nu(\alpha)$$

By Fubini

$$\begin{aligned}
\varrho(X) &= -\int_0^1 \alpha^{-1} \int_0^\alpha F^{-1}(t) dt d\nu(\alpha) \\
&= -\int_0^1 F^{-1}(t) \int_t^1 \alpha^{-1} d\nu(\alpha) dt \\
&\equiv -\int_0^1 F^{-1}(t) \gamma(t) dt
\end{aligned}$$

Approximating General Risk Measures

We can approximate general pessimistic risk measures by taking

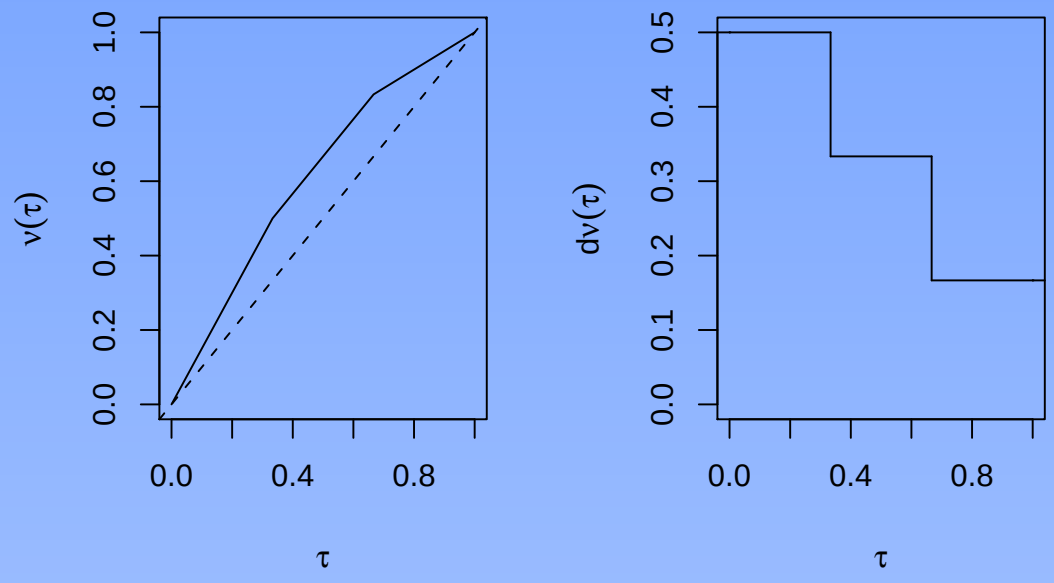
$$d\nu(t) = \sum \nu_i \delta_{\tau_i}(t)$$

where δ_τ denotes (Dirac) point mass 1 at τ Then

$$\varrho(X) = \int_0^1 F^{-1}(t)\gamma(t)dt$$

where $\gamma(t) = \sum \nu_i \tau_i^{-1} I(t < \tau_i)$ and $\nu_i > 0$, with $\sum \nu_i = 1$.

An Example



$$d\nu(t) = \frac{1}{2}\delta_{1/3}(t) + \frac{1}{3}\delta_{2/3}(t) + \frac{1}{6}\delta_1(t)$$

A Theorem

Theorem (Kusuoka (2001)) A regular risk measure is *coherent* if and only if it is *pessimistic*.

A Theorem

Theorem (Kusuoka (2001)) A regular risk measure is *coherent* if and only if it is *pessimistic*.

- Pessimistic Choquet risk measures correspond to *concave* ν, i.e., *monotone decreasing* $d\nu$.

A Theorem

Theorem (Kusuoka (2001)) A regular risk measure is *coherent* if and only if it is *pessimistic*.

- Pessimistic Choquet risk measures correspond to *concave* ν, i.e., *monotone decreasing* $d\nu$.

- Probability assessments are distorted to accentuate the probability of the least favorable events.

A Theorem

Theorem (Kusuoka (2001)) A regular risk measure is *coherent* if and only if it is *pessimistic*.

- Pessimistic Choquet risk measures correspond to *concave* ν, i.e., *monotone decreasing* $d\nu$.
- Probability assessments are distorted to accentuate the probability of the least favorable events.
- The crucial coherence requirement here is subadditivity, or submodularity, or 2-alternating in the terminology of Choquet capacities.

An Example

Samuelson (1963) describes asking a colleague at lunch whether he would be willing to make a

$$50 - 50 \text{ bet} \quad \left\langle \begin{array}{l} \text{win } 200 \\ \text{lose } 100 \end{array} \right.$$

An Example

Samuelson (1963) describes asking a colleague at lunch whether he would be willing to make a

$$50 - 50 \text{ bet} \quad \left\langle \begin{array}{l} \text{win } 200 \\ \text{lose } 100 \end{array} \right.$$

The colleague responded

"no, but I *would* be willing to make 100 such bets."

An Example

Samuelson (1963) describes asking a colleague at lunch whether he would be willing to make a

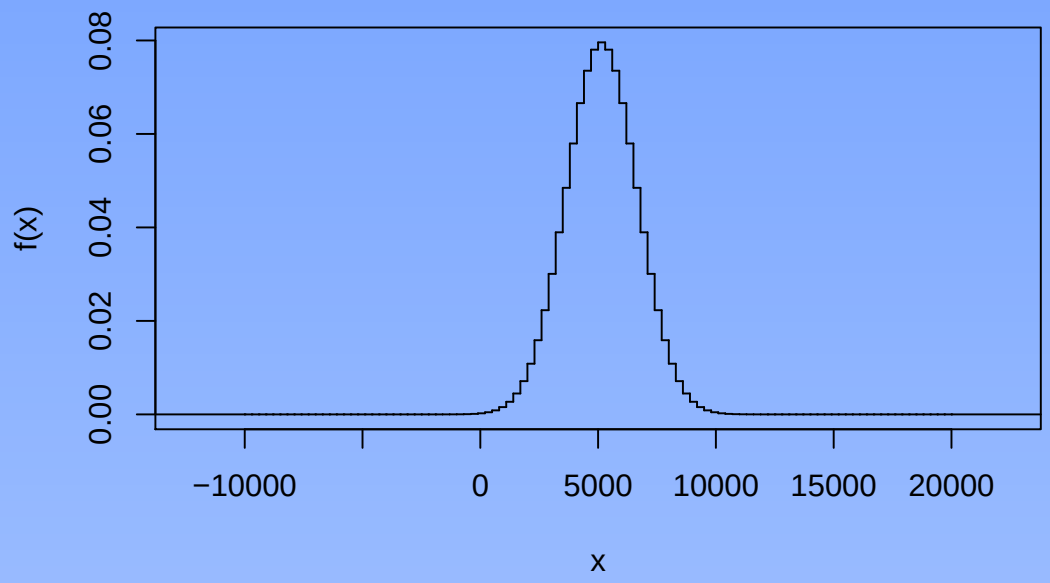
$$50 - 50 \text{ bet} \quad \left\langle \begin{array}{l} \text{win } 200 \\ \text{lose } 100 \end{array} \right.$$

The colleague responded

> "no, but I *would* be willing to make 100 such bets."

This response has been interpreted not only as reflecting a basic confusion about how to maximize expected utility but also as a fundamental misunderstanding of the law of large numbers.

Payoff Density of 100 Samuelson trials



Odds of losing money on the 100 trial bet is 1 chance in 2300.

Was Samuelson's colleague really irrational?

Suppose, for the sake of simplicity that

$$d\nu(t) = \tfrac{1}{2}\delta_{1/2}(t) + \tfrac{1}{2}\delta_1(t)$$

so for one Samuelson coin flip we have the unfavorable evaluation,

$$E_{\nu,F}(X) = \tfrac{1}{2}(-100) + \tfrac{1}{2}(50) = -25$$

but for $S = \sum_{i=1}^{100} X_i \sim \mathcal{B}in(.5, 100)$ we have the favorable evaluation,

$$\begin{aligned} E_{\nu,F}(S) &= \tfrac{1}{2}2\int_0^{1/2} F_S^{-1}(t)dt + \tfrac{1}{2}(5000) \\ &= 1704.11 + 2500 \\ &= 4204.11 \end{aligned}$$

How to be Pessimistic

Theorem Let X be a real-valued random variable with $EX = \mu < \infty$, and $\rho_\alpha(u) = u(\alpha - I(u < 0))$. Then

$$\min_{\xi \in \mathcal{R}} E\rho_\alpha(X - \xi) = \alpha\mu + \varrho_{\nu_\alpha}(X)$$

So α risk can be estimated by

$$\hat{\varrho}_{\nu_\alpha}(x) = (n\alpha)^{-1} \min_{\xi} \sum \rho_\alpha(x_i - \xi) - \hat{\mu}_n$$

How to be Pessimistic

Theorem Let X be a real-valued random variable with $EX = \mu < \infty$, and $\rho_\alpha(u) = u(\alpha - I(u < 0))$. Then

$$\min_{\xi \in \mathcal{R}} E\rho_\alpha(X - \xi) = \alpha\mu + \varrho_{\nu_\alpha}(X)$$

So α risk can be estimated by

$$\hat{\varrho}_{\nu_\alpha}(x) = (n\alpha)^{-1} \min_{\xi} \sum \rho_\alpha(x_i - \xi) - \hat{\mu}_n$$

It was inevitable: eventually everything looks like quantile regression to this guy!

Pessimistic Portfolios

Now let $X = (X_1, \ldots, X_p)$ denote a vector of potential portfolio asset returns and $Y = X^\top \pi$, the returns on the portfolio with weights π. Consider

$$\min_{\pi} \varrho_{\nu_\alpha}(Y) - \lambda \mu(Y)$$

Minimize α-risk subject to a constraint on mean return.

Pessimistic Portfolios

Now let $X = (X_1, \ldots, X_p)$ denote a vector of potential portfolio asset returns and $Y = X^\top \pi$, the returns on the portfolio with weights π. Consider

$$\min_\pi \varrho_{\nu_\alpha}(Y) - \lambda \mu(Y)$$

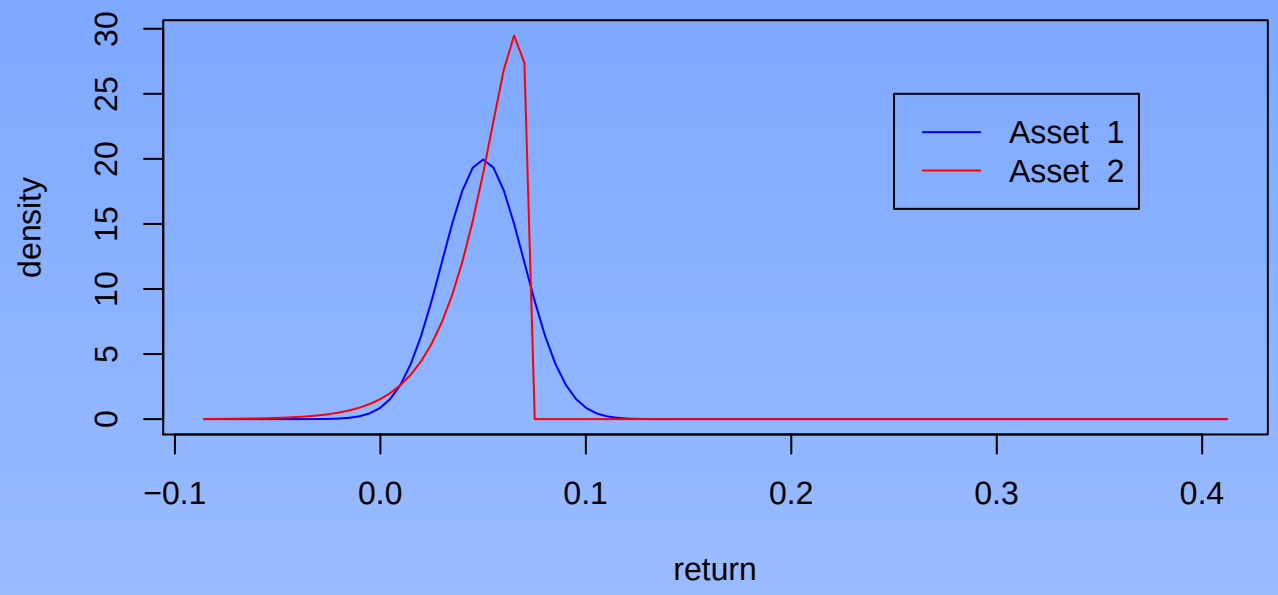
Minimize α-risk subject to a constraint on mean return.

This problem can be formulated as a quantile regression problem

$$\min_{(\beta,\xi)\in\mathcal{R}^p} \sum_{i=1}^n \rho_\alpha(x_{i1} - \sum_{j=2}^p (x_{i1} - x_{ij})\beta_j - \xi) \quad s.t. \quad \bar{x}^\top \pi(\beta) = \mu_0,$$

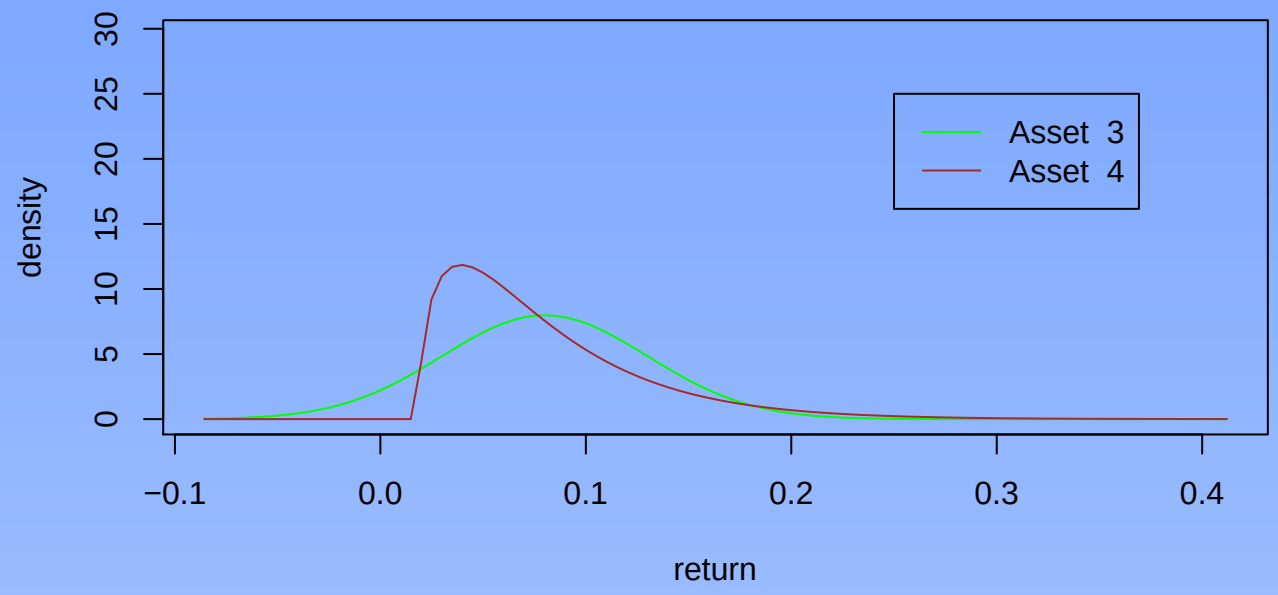
where $\pi(\beta) = (1 - \sum_{j=2}^p \beta_j, \beta^\top)^\top$.

An Example



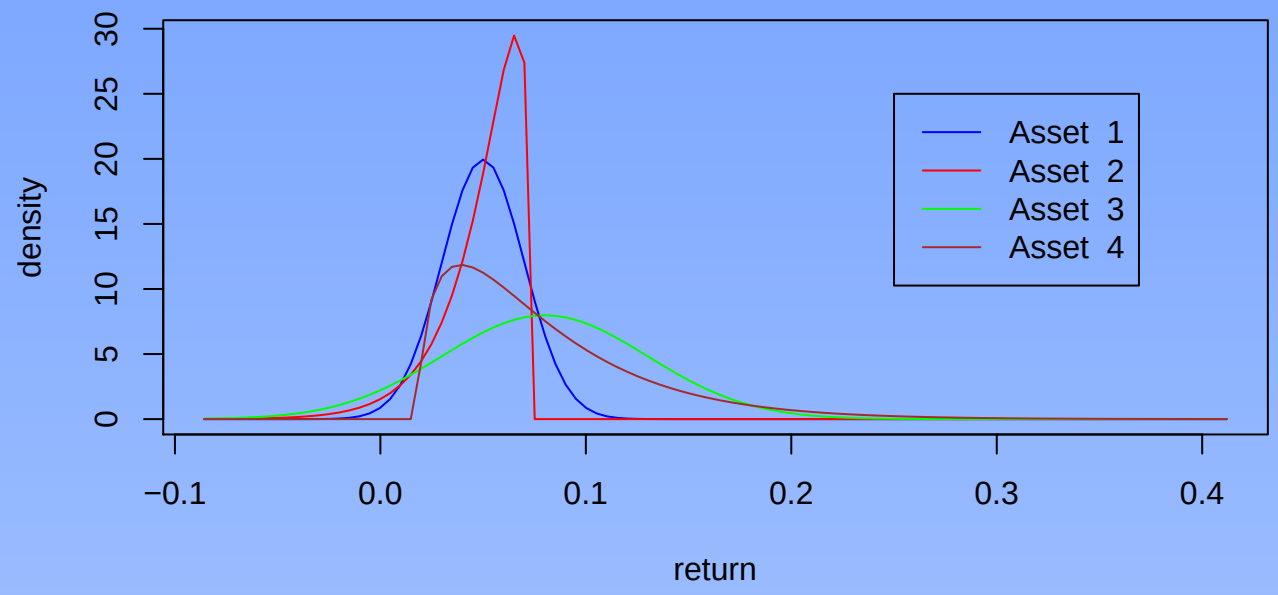
Two asset return densities with identical mean and variance.

An Example



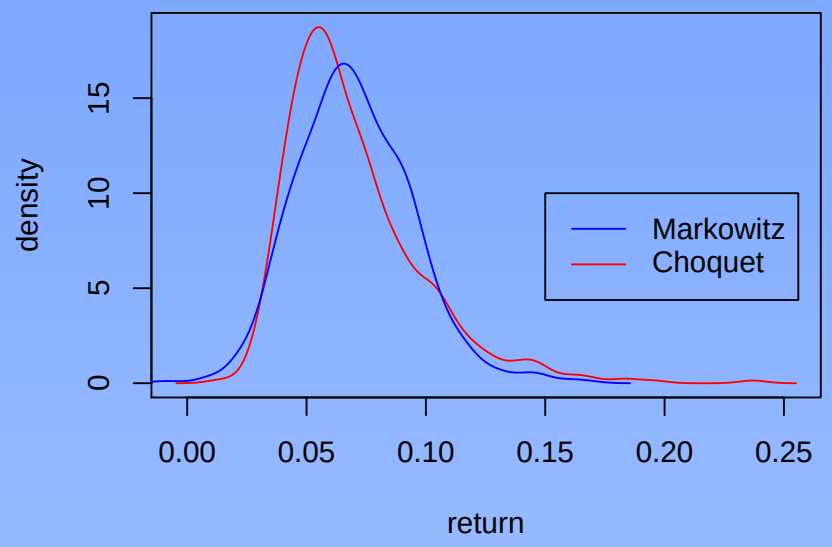
Two more asset return densities with identical mean and variance.

An Example

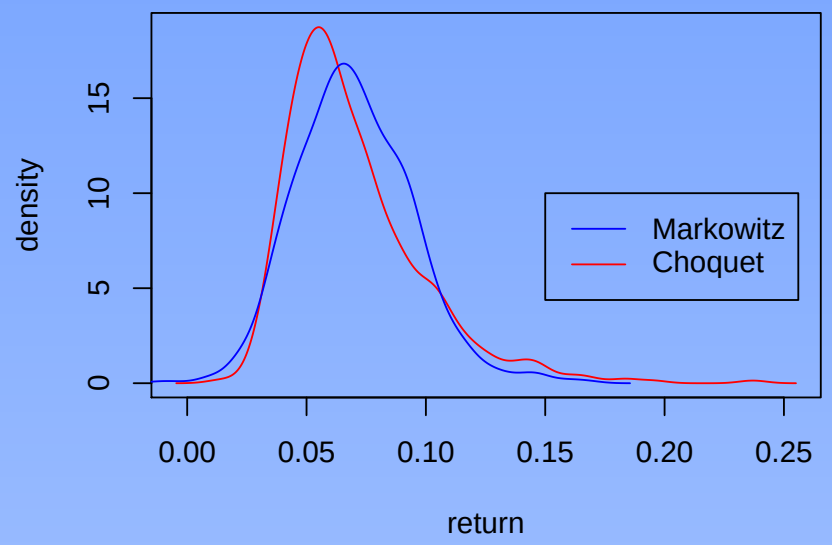


Two pairs of asset return densities with identical mean and variance.

Optimal Choquet and Markowitz Portfolio Returns



Optimal Choquet and Markowitz Portfolio Returns



Choquet portfolio has better performance in both tails than mean-variance Markowitz portfolio.

A Unified Theory of Pessimism

Any pessimistic risk measure may be approximated by

$$\varrho_\nu(X) = \sum_{k=1}^{m} \nu_k \varrho_{\nu_{\alpha_k}}(X)$$

where $\nu_k > 0$ for $k = 1, 2, ..., p$ and $\sum \nu_k = 1$.

Portfolio weights can be estimated for these risk measures by solving linear programs that are weighted sums of quantile regression problems.

Software coming soon to a CRAN archive near you!

Conclusions

- Expected Utility is unsatisfactory both as a positive, i.e., descriptive, theory of behavior and as a normative guide to behavior.

Conclusions

- Expected Utility is unsatisfactory both as a positive, i.e., descriptive, theory of behavior and as a normative guide to behavior.

- Choquet (non-additive, rank dependent) expected utility provides a simple, tractable alternative.

Conclusions

- Expected Utility is unsatisfactory both as a positive, i.e., descriptive, theory of behavior and as a normative guide to behavior.

- Choquet (non-additive, rank dependent) expected utility provides a simple, tractable alternative.

- Choquet portfolio optimization can be formulated as a quantile regression problem thus providing a tractable alternative to the dominant mean-variance approach of Markowitz (1952).